MediCare — Project Planning & Management

Graduation Project — Digital Egypt Pioneers Initiative (DEPI)

Track	Full Stack .NET Web Developer
Group	ONL5_SWD5_S3
Instructor	Eng. Margret Youssef Kamal
Team Leader	Lujain Ahmed Mohamed
Date	16 Oct 2026

1. Project Proposal

Overview

MediCare is a web-based clinic management system and the team's graduation project for the Digital Egypt Pioneers Initiative (DEPI). Patients book appointments with doctors online and receive medication reminders. Doctors manage their schedules, visits and prescriptions. Admins manage the whole system.

Objectives

#	Objective
1	Easy online booking with no double bookings
2	Digital medical records and prescriptions
3	Medication reminders for patients
4	Real-time, SMS and email notifications
5	Admin dashboard with reports

Scope

In Scope	Out of Scope
Login with 3 roles (Patient, Doctor, Admin)	Online payment
Doctor search and booking	Video consultations
Medical records and prescriptions	Mobile app
Medication schedule and reminders	Pharmacy ordering
Notifications, reports and Azure deployment	

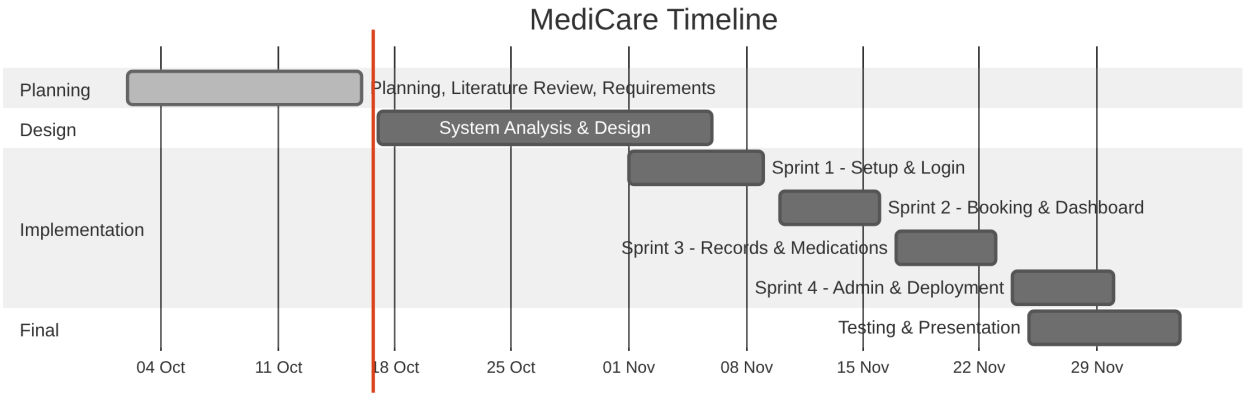
Tech Stack

ASP.NET Core MVC · Entity Framework Core · SQL Server · ASP.NET Identity · SignalR · Twilio · SendGrid · FullCalendar.js · Chart.js · Bootstrap 5 · Microsoft Azure

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2. Project Plan

Timeline



Milestones & Deliverables

Date	Deliverable
16 Oct 2026	Project Planning, Literature Review, Requirements Gathering
6 Nov 2026	System Analysis & Design
30 Nov 2026	Source code and live deployment
4 Dec 2026	Testing, reports and final presentation

Resources

Resource	Details
Team	5 members (3 backend, 2 frontend)
Tools	Visual Studio, SQL Server, GitHub, Figma
Services	Azure, Twilio, SendGrid (free or student plans)

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3. Task Assignment & Roles

Member	Role	Tasks
Lujain Ahmed Mohamed	Team Leader, Backend	Project setup, login and roles, admin backend, deployment
Ahmed Aboelela	Backend	Database, booking system, real-time notifications
Omnia Hassan	Backend	Medical records, prescriptions, medications, SMS and email
Mohamed Saleh Hussein	Frontend	Patient pages: search, booking, profile
Omnia Mostafa	Frontend	Doctor and admin dashboards, charts

4. Risk Assessment & Mitigation

Risk	Level	Mitigation
Short time	High	Weekly sprints, core features first
Code conflicts between members	Medium	Separate branches and code review
Double booking	Medium	Database check inside a transaction
SMS and email service limits	Medium	In-app notifications as a backup
Deployment problems	Medium	Deploy a test version early in Sprint 2
Medical data security	High	Role-based access, HTTPS, input validation

5. KPIs

KPI	Target
Page response time	< 2 seconds
System uptime	≥ 99%
Double bookings	0
Reminders sent on time	≥ 95%
Steps to book an appointment	≤ 4
User satisfaction during testing	≥ 4 / 5